Paul Hondola

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Education

Bachelor of Science in Computer Engineering Oct. 2023 – Jul. 2027 (expected)

Universitatea Politehnica Timişoara | GPA: 9.72/10 | Available Jul. – Sep. 2027

- Relevant Coursework: Data Structures and Algorithms, Parallel and Distributed Algorithms, Operating Systems, Computer Architecture, Formal Languages and Compilation Techniques, Software Architecture, Networking, Databases.

Experience

Software Engineer Intern Mar. 2026 – Present

Nokia | Python, pandas, MariaDB, Jenkins, GitLab, Power BI

- Halved runtime of a high-volume financial data pipeline running 8×/day (2 h → 1 h, ∼8 compute-hours saved daily) by replacing sequential row-by-row business logic with vectorized pandas operations; verified equivalence with a prod-vs-dev table diff that exposed silently dropped entries in the legacy implementation.
- Restored 3 product-management data pipelines broken by an upstream access change with a minimal hotfix (HTML → XML export parsing), then proposed and executed a ∼90-file rewrite that eliminated duplicated logic through a shared module layer and a uniform index/parser/pipeline structure per data source.

Full Stack Developer Intern Jul. 2025 – Oct. 2025

Hibyte | TypeScript, Angular, NestJS, Payload CMS, Supabase

- Built the role-based authentication layer (Supabase JWTs validated by NestJS guards across admin, staff and customer roles) and backend routes for a game-center management proof of concept in a 5-person team.
- Configured 7 Payload CMS collections for non-technical admins and maintained Supabase migrations through continuous schema changes.

Reverse Engineering and Malware Analysis Trainee Apr. 2025 – Jun. 2025

Bitdefender × LigaAC | C, Python, x86 Assembly, IDA Pro, Frida

- Reversed a file-encryption routine from x86 disassembly in IDA Pro and implemented working decryptors in C for 5 file formats, plus a keygen reproducing a licence algorithm from recovered lookup tables and hash logic.
- Unpacked dynamically loaded Android DEX payloads with Frida hooks on DexClassLoader/DexFile and built Python/Java tools to decrypt locker-app data across 8 reverse-engineering labs on purpose-built binaries.

Projects

Groundly (Bachelor Thesis) | *Python, FastMCP, SQLite, bge-m3, GraphRAG* Jul. 2026 – Present

- Built an MCP server (tested with Claude Code and Codex) that indexes course materials into portable single-file knowledge bases, serving hybrid retrieval (bge-m3 dense + learned sparse + BM25, reciprocal rank fusion, cross-encoder reranking) with zero API keys.
- Evaluated 3 GraphRAG retrieval strategies against a reranked hybrid baseline on 124 gold questions across 2 course corpora; after GraphRAG failed to outperform it, removed query-time GraphRAG and the external-model answer pipeline for an MCP-first, zero-key architecture.
- Cut peak indexing memory from >16 GB to a flat 5.05 GB independent of document size by profiling with memray and streaming fp16 embeddings in bounded batches into SQLite.

Sentinela (5-person team) | *Python, FastAPI, PostgreSQL, Kotlin, SvelteKit, Twilio, Railway* Oct. 2026 – Present

- Building a safety-check system for seniors living alone, live on Railway staging and production: Twilio check-in calls detect alarm phrases and open incidents for a dispatch team's Kanban board (open / in progress / resolved), refreshed every 15 s.
- Medics register seniors via a four-step Android flow recording consent as an auditable history (withdrawals append, never overwrite), backed by a FastAPI/PostgreSQL service with 194 tests.

gemm-bench | *Rust, Rayon, std::simd, SQLite, Svelte* Sep. 2026 – Present

- Built 9 Rust CPU GEMM kernels (cache tiling, Rayon work-stealing, static thread pools, a BLIS-style packed micro-kernel with `std::simd` register blocking), benchmarked against Apple AMX and Metal GPU backends at 5 precisions.
- Reached 679 GFLOPS f32 with the parallel packed micro-kernel (87% of the M1 Pro's theoretical NEON peak), vs. 1.9–2.3 TFLOPS on AMX and 3.5 TFLOPS on the GPU.
- Gathered results from 8 machines and 4 contributors into strict-schema SQLite databases, validated in CI on Ubuntu, Windows and macOS (147 Rust tests) and charted in a live dashboard.

Vibecodium | 1st Place, iTEC 2026 | *TypeScript, React, Bun, Supabase, Yjs, Docker* Mar. 2026 – Jun. 2026

- Co-led a 5-person team building a real-time collaborative IDE (Yjs CRDT), owning the workspace UI and REST APIs, and built a Docker code-execution sandbox for 4 languages with 2 GB memory caps, no network access and a 3 s timeout.

Distributed Image Processing (Parallel Algorithms coursework) | *C, MPI, OpenMP* Dec. 2025 – Feb. 2026

- Parallelized 2D convolution on images up to 16800×12600 px across 5 execution models (serial, OpenMP, MPI message passing, shared-file I/O, master-worker task pool), reaching a median 6.8× speedup with 8 MPI processes.

Technical Skills

Languages: Python, C, C++, Rust, Java, TypeScript, SQL, Bash
Parallel and HPC: MPI, OpenMP, Rayon, POSIX Threads, SIMD/NEON, Cache Optimization
Systems and Tools: Linux, Docker, Jenkins, GitHub Actions, GitLab CI, Railway, Meson, Cargo, Maven, IDA Pro, Frida
Data: pandas, NumPy, DuckDB, Parquet, Power BI, PostgreSQL, MariaDB, SQLite
Web: React, Angular, Svelte, NestJS, FastAPI, Hono, Supabase
Spoken Languages: English (C1/Advanced), Romanian (Native)